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Climate transition risk premium in equity markets: Downside asymmetry and regime dependence across Europe, the United States, and the United Kingdom

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ABSTRACT

This paper examines the pricing of climate transition risk in equity markets through the return differential between clean energy and brown indices across Europe, the United States, and the United Kingdom. Using daily data, the analysis combines OLS regressions, factor-based specifications, rolling-window estimates, downside beta analysis, quantile regressions, tail-risk measures, and event-based tests, complemented by a green bond robustness check. The results show that the transition-related equity premium is not a stable unconditional spread, but rather a conditional and time-varying phenomenon. Its magnitude varies across regions, responds to market, energy, and carbon-pricing conditions, and becomes more pronounced during periods of stress and in the tails of the return distribution. The factor-based analysis identifies industrial and real-economy conditions as the most robust additional set of determinants, while financial stress plays a more episodic role. Downside beta and quantile evidence reveal substantial asymmetry in the behavior of the clean-minus-brown spread, with the United Kingdom exhibiting the strongest tail sensitivity. Overall, the findings indicate that transition risk is reflected in equity prices, but in an asymmetric, state-dependent, and heterogeneous manner across markets.

1. Introduction

As the effects of global climate change become more severe, hazards such as extreme weather events and rising sea levels, driven by greenhouse gas emissions, pose growing risks to social stability, environmental sustainability, and economic growth (Gan et al., 2026; Luangaram et al., 2026). In this context, the transition toward a low-carbon economy represents one of the defining structural shifts of the coming decades (Di Vaio et al., 2025). As governments implement climate policies, investors reassess their exposure to carbon-intensive assets, and technological change reshapes competitive dynamics across sectors, the financial implications of this transition have become increasingly important in asset pricing research (Ardia et al., 2023; Ceccarelli et al., 2024; Clémenton, 2023; Călin, Lupu, Dumitrescu, et al., 2026; Tumewang et al., 2025). Understanding how these forces are reflected in financial markets is relevant not only for portfolio allocation and risk management, but also for broader questions concerning the

efficiency of capital reallocation and the stability of the financial system during large-scale economic transitions (Iqbal et al., 2025; Javed & Bhutta, 2026; Luangaram et al., 2026; Ojea-Ferreiro et al., 2024).

A growing body of research has documented how climate-related risks—both physical and transition-related—affect asset valuations. Physical climate risk refers to the economic damage caused by extreme weather events and long-term changes in climate, while transition risk arises from policy, technological, and preference shifts associated with decarbonization (Caraiani et al., 2026; Huang, Cheng, & Zhang, 2025; Zhou & Ma, 2025). Although physical risk has received considerable attention in empirical studies of corporate performance and real estate markets (Bartolini et al., 2025; Huang et al., 2025; Xue et al., 2026), the pricing of transition risk in equity markets remains less well understood, particularly across jurisdictions and under different market conditions.

Transition risk is typically expected to manifest through the differential pricing of “green” and “brown” assets—that is, through a wedge between the returns on low-carbon and carbon-intensive equities. If

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investors anticipate regulatory costs, exposure to stranded assets, or reputational penalties for fossil-fuel-dependent firms, brown assets should command a risk premium, implying lower valuations and higher expected returns relative to clean-energy equities (Chan & Wan, 2026; Hambel & van der Ploeg, 2025). Conversely, if green assets benefit from policy support, technological momentum, or investor preferences for sustainable portfolios, they may command a “green premium” that reflects factors such as lower liquidity, regulatory uncertainty, or concentration risk (Hsu et al., 2023; Peng et al., 2026; Tang et al., 2025). The net effect—whether reflected in persistent green outperformance or a brown discount—remains an empirical question, one that may vary across markets and over time.

Existing empirical evidence on transition-risk pricing in equity markets is mixed. Some studies find that firms with higher carbon emissions or greater exposure to fossil fuels trade at a discount relative to their low-carbon peers (Birk et al., 2026; de Dios-Aluja et al., 2021; Duong, 2025), while others report limited or inconsistent pricing effects (Schoeffel et al., 2026; Tang & Zhang, 2020). This heterogeneity may reflect differences in sample periods, market coverage, or empirical design. More importantly, it may suggest that transition risk is not priced as a stable, unconditional premium, but instead varies systematically with market conditions, policy regimes, and exposure to macroeconomic or energy-market shocks (Duong, 2025; Zhu et al., 2026).

In parallel, a separate strand of the literature has emerged on green bond pricing, documenting the existence of a “greenium”—the yield differential between green and conventional bonds—across several markets (Eskildsen et al., 2024; Berdiev, 2025; Fricke & Meinerting, 2025; D’Amico et al., 2026).

Although green bonds and green equities represent distinct asset classes with different risk profiles and investor bases, both segments reflect investor demand for climate-aligned investments. Comparing equity-based transition spreads with bond-based greenium measures can therefore provide insights into the consistency of transition-risk pricing across different segments of the capital structure (Bartolini et al., 2025; Hossain et al., 2026).

Against this background, this paper investigates the pricing of climate transition risk in equity markets by examining the return differential between clean energy and brown equity indices across three major developed markets: Europe, the United States, and the United Kingdom. The analysis covers the period from 2010 to 2026, providing a sufficiently long sample to capture multiple policy shocks, energy-market disruptions, and episodes of macroeconomic stress, including the COVID-19 pandemic and the war in Ukraine. Rather than testing for a single, stable transition premium, the paper adopts a conditional perspective that emphasizes heterogeneity across markets, asymmetry during downside episodes, and regime-dependent behavior around major events.

2. Main contributions

This paper contributes to the literature on climate finance and transition-risk pricing along several dimensions. First, it examines transition risk through the return differential between clean energy and brown equity indices, thereby focusing directly on the relative pricing of low-carbon and carbon-intensive assets. This spread-based approach provides a targeted measure of transition-related pricing by capturing the relative market valuation of green and brown assets rather than considering either segment in isolation.

Second, the paper provides a comparative cross-market perspective by examining Europe, the United States, and the United Kingdom within a unified empirical framework. This design makes it possible to assess whether transition-risk pricing follows a common pattern across developed markets or varies systematically with institutional settings, market composition, and exposure to energy and policy shocks. In doing so, the study extends the existing literature by highlighting the cross-market heterogeneity of transition-related pricing.

Third, and most importantly, the paper moves beyond average-return analysis by showing that transition-risk pricing is fundamentally conditional. The empirical strategy combines baseline OLS regressions with downside beta estimation, quantile regressions, tail-risk metrics, and event-based evidence to capture asymmetry and regime dependence. This allows the analysis to identify not only whether transition risk is priced, but also when and under what market conditions it becomes most salient. In this sense, the paper contributes to the literature by demonstrating that the transition premium is better understood as a state-dependent and tail-sensitive phenomenon rather than as a stable unconditional effect.

Fourth, the paper jointly evaluates the role of market-wide conditions, oil-price dynamics, and carbon-pricing shocks in explaining the transition spread. This helps distinguish between broad financial-market influences and more specific transition-related channels. The results suggest that carbon-pricing dynamics carry a more systematic signal than oil-price movements, particularly in Europe and the United States, thereby contributing to the debate on how climate-policy variables are reflected in financial valuations.

Finally, the paper extends the analysis beyond equities through a bond-market robustness exercise based on the greenium. Although the cross-asset evidence is mixed, this extension provides a useful bridge between the equity pricing of transition risk and the broader sustainable finance literature. It also addresses the question of whether transition-related pricing signals are confined to one segment of financial markets or transmitted more broadly across asset classes.

Taken together, these contributions show that climate transition risk is reflected in equity prices, but not in the form of a simple, stable premium. Rather, it is priced in a manner that is asymmetric, regime-dependent, and heterogeneous across markets. By documenting these features in a comparative setting, the paper provides a more nuanced account of how the low-carbon transition is incorporated into financial valuations.

3. Research hypotheses

Building on the theoretical framework and the emerging empirical literature on transition risk, this paper investigates whether climate-transition considerations are reflected in cross-sectional and time-varying pricing differentials between green and brown assets. The analysis focuses on equity-market evidence across Europe, the United States, and the United Kingdom, while also considering the role of energy prices, carbon pricing, market-wide conditions, and stress episodes. In line with these objectives, the paper tests the following hypotheses.

H1. Existence of a transition premium. A transition-related equity premium exists and is reflected in systematic return differentials between green and brown equity indices. If transition risk is priced in equity markets, the relative performance of clean-energy and carbon-intensive assets should not be random, but should instead display a persistent and economically meaningful spread over time.

H2. Cross-market heterogeneity. The transition-related equity premium differs significantly across the European, US, and UK markets. This hypothesis reflects the possibility that institutional structures, sectoral composition, investor preferences, and regulatory environments shape the pricing of transition risk differently across jurisdictions.

H3. Role of energy and carbon pricing. The transition spread is significantly influenced by energy prices and carbon-pricing dynamics. Since the valuation of green and brown assets is closely linked to fossil-fuel dependence and climate-policy exposure, changes in oil and carbon prices are expected to affect the relative pricing of these asset classes.

H4. Market-wide conditions. The transition spread is sensitive to overall market conditions. In addition to transition-specific drivers, broad market movements may alter investors' relative demand for green and brown assets, implying that the spread should co-move with aggregate

market returns.

H5. Regime dependence. The transition-related equity premium varies across market regimes, particularly during periods of financial stress. This hypothesis captures the idea that transition-risk pricing is not constant over time, but becomes more pronounced during episodes of macroeconomic disruption, elevated uncertainty, or market-wide repricing.

H6. Downside asymmetry. The transition spread exhibits asymmetric behavior during downside market conditions. In particular, the relative performance of green and brown assets may differ more markedly during negative market states than during normal periods, implying that downside episodes contain distinct information about the pricing of transition risk.

H7. Consistency with bond markets. As a robustness test, the equity-based transition spread is expected to be broadly consistent with pricing differentials observed in green bond markets over the available sample period. If transition risk is priced across asset classes, the signals extracted from equity markets should display at least partial alignment with the greenium observed in fixed-income markets.

Taken together, these hypotheses allow the paper to assess not only whether a transition-related premium exists, but also whether it varies across markets, depends on economically relevant pricing factors, intensifies under stress, and remains broadly consistent across asset classes. The following section describes the data and empirical methodology used to test these hypotheses. Section 3 then presents the empirical results, including baseline regressions, factor-based determinant analysis, time-varying evidence, downside and tail-risk evidence, event-based analysis, and bond-market robustness checks. The final section concludes.

4. Data and methodology

4.1. Data

The empirical analysis relies on two daily-frequency datasets. The first dataset, hereafter referred to as the long sample, covers the main equity-market analysis over the 2010–2026 period. The second dataset, hereafter referred to as the short sample, is used for the bond-market robustness exercise and covers the shorter 2023–2026 period, reflecting data availability for green and conventional bond indices.

The long sample includes three groups of equity indices, one for each geographic market: Europe, the United States, and the United Kingdom. For each market, the dataset contains a clean energy equity index, a brown equity index, and a broad market benchmark. In addition, the long sample includes two external transition-related factors: Brent crude oil prices and EU carbon allowance prices. The short sample contains the same equity and factor series, supplemented by green bond and conventional corporate bond indices for each market.

For Europe, the clean equity leg is proxied by the S&P Global Clean Energy index, the brown leg by the STOXX Europe 600 Oil and Gas index, and the aggregate market by the STOXX Europe 600 index. For the United States, the clean leg is represented by the S&P Global Clean Energy index, the brown leg by the S&P 500 Energy sector index, and the market benchmark by the S&P 500 Composite. For the United Kingdom, the clean leg is again proxied by the S&P Global Clean Energy index, the brown leg by the FTSE 350 Oil, Gas and Coal index, and the market benchmark by the FTSE 100 index. Transition-related external factors are measured using Brent spot crude oil prices and ICE EUA carbon futures prices. In the robustness analysis, bond-market pricing is captured using iBoxx green bond indices and broad corporate bond indices for the EUR, USD, and GBP markets.

The dataset also includes sector-level variables used to construct additional determinant factors. The financial/stress factor is based on European and US financial-sector indices and the VSTOXX volatility

index, while the industrial/real-economy factor is based on European and US industrials, materials, and utilities sector indices. These variables are included to capture broader financial and sectoral conditions that may affect the relative performance of clean and brown equity exposures beyond the baseline market, Brent, and carbon-pricing controls.

All series are organized at a daily frequency and transformed into continuously compounded returns using log differences.

4.2. Variable construction

For each price series $P_{i,t}$, daily returns are computed as log returns:

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right)$$

where $r_{i,t}$ denotes the return on asset i at time t .

The central dependent variable is the transition spread, defined as the difference between the returns on the clean energy and brown equity indices within each market:

$$Spread_{m,t} = r_{m,t}^{clean} - r_{m,t}^{brown}$$

where $m \in \{EU, US, UK\}$.

$$Spread_{EU,t} = r_{EU,t}^{CE} - r_{EU,t}^{Brown}$$

$$Spread_{US,t} = r_{US,t}^{CE} - r_{US,t}^{Brown}$$

$$Spread_{UK,t} = r_{UK,t}^{CE} - r_{UK,t}^{Brown}$$

A positive value of the spread implies that clean energy equities outperform brown equities on that day, while a negative value implies the opposite.

The explanatory variables used in the main regressions are the corresponding broad market return, Brent oil return, and carbon price return:

$$r_{m,t}^{MKT}, r_t^{Brent}, r_t^{Carbon}$$

For the bond-market robustness exercise, a bond greenium variable is constructed as the return differential between the green bond and conventional corporate bond indices:

$$Greenium_{m,t} = r_{m,t}^{GB} - r_{m,t}^{Corp}$$

where $r_{m,t}^{GB}$ is the return on the green bond index and $r_{m,t}^{Corp}$ is the return on the conventional corporate bond index for market m .

4.3. Descriptive analysis

The empirical analysis begins with descriptive statistics for the transition spreads and the main explanatory variables. For each series, the following summary measures are computed: mean, standard deviation, minimum, lower quartile, median, upper quartile, maximum, skewness, and kurtosis. In addition, normality is assessed using the Jarque-Bera test.

This first step serves two purposes. First, it provides an initial assessment of whether the transition spread exhibits a non-zero average, which is relevant to H1. Second, it offers preliminary evidence on cross-market differences in distributional properties, which is relevant to H2. The higher-order moments and normality tests also motivate the use of tail-sensitive methods in later stages of the analysis.

4.4. Baseline regression specification

To examine the determinants of the transition spread, the paper estimates a baseline linear model separately for each market. The specification is:

$$\text{Spread}_{m,t} = \alpha_m + \beta_{1,m} r_{m,t}^{\text{MKT}} + \beta_{2,m} r_t^{\text{Brent}} + \beta_{3,m} r_t^{\text{Carbon}} + \varepsilon_{m,t}$$

where:

- $\text{Spread}_{m,t}$ is the transition spread in market m ,
- $r_{m,t}^{\text{MKT}}$ is the return on the aggregate market index,
- r_t^{Brent} is the daily Brent oil return,
- r_t^{Carbon} is the daily carbon price return,
- $\varepsilon_{m,t}$ is the disturbance term.

The model is estimated using ordinary least squares (OLS), with HC3 heteroskedasticity-robust standard errors. This specification is designed to test whether the transition spread is significantly related to energy-price shocks and carbon-pricing dynamics, as proposed in H3, and whether it co-moves with overall market conditions, as proposed in H4.

The coefficient $\beta_{1,m}$ captures the sensitivity of the spread to general market movements. The coefficient $\beta_{2,m}$ measures the effect of oil-price dynamics, while $\beta_{3,m}$ measures the effect of carbon-pricing shocks. Cross-market variation in these coefficients also provides additional evidence on heterogeneity across regimes.

4.5. Factor-based determinant analysis

To address the role of broader determinants, the baseline specification is extended with two factor-based blocks. The first is a financial/stress factor, designed to capture broad financial-sector and stress-related conditions. It is constructed as the first principal component of European financial-sector returns, US financial-sector returns, and changes in the VSTOXX volatility index. The factor is oriented so that higher values correspond to greater financial stress. The second is an industrial/real-economy factor, constructed as the first principal component of European and US industrials, materials, and utilities returns. This factor captures common variation in sectoral real-economy conditions that may affect the relative performance of clean and brown equity exposures.

The extended determinant regressions add these factors to the baseline market, Brent, and carbon-pricing controls. The analysis is implemented through nested specifications to assess the incremental explanatory power of each factor block. In addition, to verify that the factor-based results are not simply capturing broad market movements, the financial/stress and industrial factors are orthogonalized with respect to the corresponding regional market return. These orthogonalized factors are then included in the regressions together with the baseline controls. This robustness step allows the analysis to assess whether financial and industrial conditions contain information beyond aggregate market exposure.

Finally, the factor-based analysis is complemented by two forms of time-varying evidence. First, 252-trading-day rolling regressions are estimated to examine whether the contribution of the industrial factor changes over time. Second, subsample regressions are estimated for the pre-COVID, COVID/stress, energy-crisis, and post-crisis periods. These tests assess whether the determinants of the transition spread differ across market regimes.

4.6. Downside beta analysis

To assess whether transition-risk pricing behaves differently under adverse market conditions, the analysis computes a downside beta for each market. This is done by restricting the sample to days on which the corresponding broad market return is negative:

$$r_{m,t}^{\text{MKT}} < 0$$

Within this conditional sample, the following regression is estimated:

$$\text{Spread}_{m,t} = \alpha_m^D + \beta_m^D r_{m,t}^{\text{MKT}} + u_{m,t}$$

where β_m^D is the downside beta.

A negative and statistically significant downside beta indicates that the relationship between market returns and the transition spread changes under adverse market states. Since the estimation is restricted to days with negative market returns, a negative coefficient implies that larger market declines are associated with an increase in the clean-minus-brown spread. This provides a direct test of H6 and offers a first formal approach to state-dependent asymmetry.

4.7. Quantile regression

To move beyond average effects and investigate whether the determinants of the transition spread vary across its distribution, the paper estimates quantile regressions at several points of the conditional distribution, specifically at the 10th, 25th, 50th, 75th, and 90th percentiles.

For each quantile τ , the estimated model is:

$$Q_\tau(\text{Spread}_{m,t} | X_{m,t}) = \alpha_{m,\tau} + \beta_{1,m,\tau} r_{m,t}^{\text{MKT}} + \beta_{2,m,\tau} r_t^{\text{Brent}} + \beta_{3,m,\tau} r_t^{\text{Carbon}}$$

where $Q_\tau(\text{Spread}_{m,t} | X_{m,t})$ denotes the conditional quantile of the spread given the regressors.

This framework allows the marginal effects of market returns, oil prices, and carbon prices to vary across the lower tail, the center, and the upper tail of the spread distribution. In substantive terms, quantile regressions make it possible to determine whether transition-risk pricing is more pronounced in downside realizations than under normal or favorable conditions. This is particularly important for H6, and more broadly for H5, since regime dependence is often reflected in the tails rather than in the mean.

4.8. Tail-risk measures

To further characterize the downside properties of the transition spread, the paper computes historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) at the 95% and 99% confidence levels. This tail-risk perspective is consistent with recent evidence showing that severe drawdowns, volatility dynamics, and tail behavior provide information on asset-market vulnerability that is not captured by average-return measures alone (Călin, Lupu, Iancu, & Dumitrescu, 2026).

The historical VaR at confidence level c is defined as the empirical quantile:

$$\text{VaR}_c = Q_{1-c}(\text{Spread}_{m,t})$$

The corresponding CVaR, also known as expected shortfall, is defined as the average spread conditional on the spread falling below the VaR threshold:

$$\text{CVaR}_c = E(\text{Spread}_{m,t} | \text{Spread}_{m,t} \leq \text{VaR}_c)$$

These measures capture the severity of extreme downside movements in the spread and complement the quantile-based analysis. More negative VaR and CVaR values indicate larger losses under adverse market conditions. This step is directly relevant for assessing the downside asymmetry and tail sensitivity described in H6.

4.9. Event-based regime analysis

To study regime dependence more explicitly, the paper conducts an event-based analysis around major macro-financial and geopolitical shocks. The selected events are the COVID-19 outbreak, the Russian invasion of Ukraine, the 2022 energy crisis peak, and the COP26 climate summit.

For each event, an event window is centered around the event date and the spread is accumulated over time. Let t_0 denote the event date.

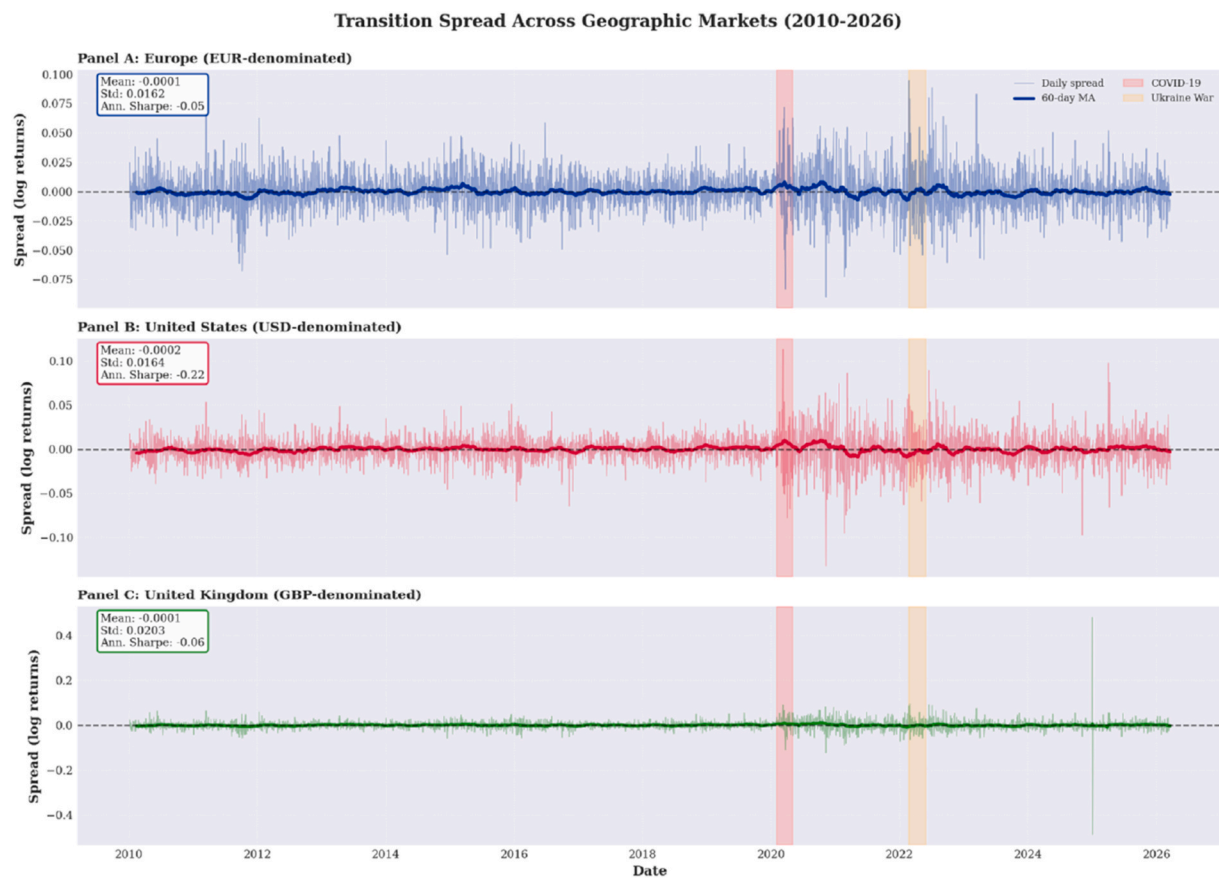


Fig. 1. Transition Spread Across Geographic Markets (2010 – 2026)

Source: Authors' computation.

For each day k in the event window, the cumulative spread is computed as:

$$CS_m(k) = \sum_{j=-w}^k Spread_{m,t_0+j}$$

where w denotes the pre-event window length.

The analysis reports cumulative spread dynamics around each event and summarizes the 30-day post-event cumulative spread. Negative cumulative values indicate that brown assets outperformed green assets after the event, whereas positive values indicate the reverse.

Strictly speaking, this procedure measures cumulative spread behavior around events rather than abnormal returns relative to an external pricing benchmark. Nevertheless, it provides a transparent way to assess whether transition-risk pricing differs across regimes and major shocks, thereby addressing H5.

4.10. Bond-market robustness analysis

To assess whether the transition-related pricing signal identified in equity markets is consistent with bond-market pricing, the paper uses the short sample to construct bond greenium measures for each market. The greenium is defined as:

$$Greenium_{m,t} = r_{m,t}^{GB} - r_{m,t}^{Corp}$$

The relationship between equity and bond pricing is then studied in two ways.

First, the contemporaneous correlation between the equity transition spread and the bond greenium is computed:

$$\rho_m = \text{Corr}(Spread_{m,t}, Greenium_{m,t})$$

Second, the following validation regression is estimated:

$$Greenium_{m,t} = \alpha_m + \gamma_m Spread_{m,t} + \eta_{m,t}$$

where γ_m measures the extent to which bond-market green pricing comoves with the equity-based transition spread.

A positive and statistically meaningful relationship would suggest that transition-related pricing is not confined to equity markets, but extends more broadly across asset classes. This constitutes the robustness test for H7. Because the short sample is more limited in length and the bond and equity measures are not constructed identically, the robustness evidence is interpreted cautiously.

4.11. Empirical mapping to hypotheses

The empirical strategy is organized around the seven hypotheses introduced earlier. H1 and H2 are first explored through descriptive statistics and visual inspection of the transition spreads across markets. H3 and H4 are evaluated using the baseline OLS regressions linking the spread to market returns, oil prices, and carbon prices. The factor-based determinant analysis further extends the empirical design by examining whether financial/stress and industrial/real-economy conditions provide additional explanatory power beyond the baseline controls. H5 is examined through event-based analysis, rolling-window estimates, subsample regressions, and, more broadly, methods that allow the spread to vary across stressed environments. H6 is tested through downside beta estimation, quantile regressions, and tail-risk measures. Finally, H7 is assessed through the bond-market robustness analysis, which compares the equity transition spread with the green bond return differential over the shorter common sample.

Table 1

Baseline determinants of the transition spread.

Variable	Europe	United States	United Kingdom
Constant	−0.0001 (0.0002)	−0.0002 (0.0002)	−0.0000 (0.0003)
Market return	−0.1943*** (0.0317)	−0.0366 (0.0337)	−0.3791*** (0.0462)
Brent return	−0.1134*** (0.0163)	−0.2483*** (0.0233)	−0.1199*** (0.0196)
Carbon return	−0.0135* (0.0082)	−0.0116 (0.0089)	−0.0065 (0.0098)
Observations	4230	4230	4230
Adj. R ²	0.058	0.111	0.061

Notes: The dependent variable is the transition spread, defined as clean equity returns minus brown equity returns. The table reports OLS estimates with HC3 robust standard errors in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Source: Authors' computation

5. Results

5.1. Data and descriptive statistics

Table A2.1 reports the summary statistics for the daily transition spreads and the main explanatory variables over the full sample. Across the three equity markets, the average transition spread, defined as the return differential between clean energy and brown equity indices, is economically small and close to zero in unconditional terms. This suggests that the transition-related premium is not expressed as a large and persistent average daily return differential throughout the sample. At the same time, the spreads display non-zero variation and distinct time-series behavior, indicating that transition-related pricing differences do exist, although they appear to be conditional and state-dependent rather than constant. In this sense, Table A2.1 provides only limited unconditional support for H1, while also motivating the more detailed conditional analyses developed in the subsequent sections.

The descriptive evidence also points to meaningful cross-market heterogeneity, in line with H2. While the European and US transition spreads display relatively similar levels of dispersion, the UK spread stands out as the most volatile and the most exposed to extreme observations. In particular, the UK series exhibits substantially higher kurtosis and a much larger maximum realization, suggesting that the pricing of transition risk is less stable and more episodic in the UK market than in the European and US cases. These differences indicate that the transition premium is not uniform across jurisdictions, but instead reflects market-specific structures, sector composition, and exposure to energy-related shocks.

Fig. 1 plots the daily transition spread for Europe, the United States, and the United Kingdom over the 2010–2026 sample. Across all three markets, the raw daily spread fluctuates closely around zero, while the 60-day moving average remains relatively close to the horizontal zero line for much of the sample. This visual evidence reinforces the conclusion that the transition-related equity premium is not expressed as a large, persistent, one-directional return differential in unconditional terms. Instead, the premium appears modest on average and strongly time-varying, which is consistent with the descriptive evidence reported in Table A2.1. In relation to H1, Fig. 1 therefore points to the existence of a transition spread, but one that is episodic and conditional rather than stable over time.

At the same time, Fig. 1 reveals important cross-market heterogeneity, providing visual support for H2. The European and US spreads display relatively similar dynamics, with frequent short-lived deviations

Table 2

Incremental explanatory power of determinant factor blocks.

Market	Baseline Adj. R ²	+ Financial/stress Adj. R ²	+ Industrial Adj. R ²	Full Adj. R ²	Delta Adj. R ² financial	Delta Adj. R ² industrial	Joint p-value
Europe	0.058	0.067	0.135	0.135	0.009	0.077	<0.001
United States	0.111	0.115	0.127	0.128	0.005	0.016	<0.001
United Kingdom	0.061	0.089	0.146	0.146	0.028	0.085	<0.001

Source: Authors' computation

around zero and moderate volatility over most of the sample. By contrast, the UK spread appears more fragile, with visibly larger spikes and more pronounced extreme observations, especially in the later part of the sample. This pattern suggests that transition-related pricing is not homogeneous across markets, but instead depends on market-specific structures, sector composition, and exposure to energy-related shocks. The UK market appears particularly sensitive to episodes of repricing, reinforcing the idea that transition risk is transmitted differently across jurisdictions.

A central feature of Fig. 1 is the marked increase in spread volatility around major stress episodes, particularly during the COVID-19 outbreak and the period following the Russian invasion of Ukraine. In all three panels, the spread becomes substantially more volatile during these windows, and short-run deviations from zero become both larger and more frequent. This behavior is especially relevant for H5, as it suggests that the transition premium is regime-dependent and intensifies during periods of macro-financial disruption, energy-market stress, and heightened uncertainty. Rather than moving smoothly through time, the spread appears to widen or compress abruptly when markets enter stress regimes, indicating that transition-related pricing is closely tied to broader shock episodes.

The figure also provides descriptive motivation for the downside and tail-risk analyses used to assess H6. In the US and UK panels in particular, stress episodes are associated with more pronounced negative spikes in the spread, implying periods in which brown assets outperform clean energy stocks. This visual pattern is consistent with the possibility that fossil-fuel-related firms may provide relative resilience during adverse market environments, especially when stress is linked to energy security, commodity price shocks, or inflationary pressure. Since Fig. 1 is descriptive, however, it does not identify downside asymmetry formally. The formal evidence is therefore provided by the downside beta, quantile-regression, and tail-risk analyses that follow.

The distributional diagnostics reported in Figure A2.2 further support this empirical strategy. The transition spreads depart systematically from normality across all three markets. Although the distributions are centered close to zero, they are more sharply peaked and display heavier tails than the Gaussian benchmark, indicating that extreme realizations occur more frequently than would be expected under a normal process. These deviations are confirmed by the Jarque-Bera statistics and the Q-Q plots, which show that the empirical quantiles diverge markedly from the fitted normal reference line in the tails.

These distributional features are economically important. Excess kurtosis is evident across all markets and is particularly pronounced in the United Kingdom, where the spread distribution exhibits the strongest tail departures. This suggests that transition-risk pricing is characterized less by a simple average shift in returns and more by episodic bursts of repricing concentrated in turbulent periods. From a methodological perspective, this supports the use of conditional and tail-sensitive methods. Inference based solely on conditional means would risk overlooking the most economically relevant features of the data. The evidence therefore motivates the use of quantile regressions, downside beta estimation, and tail-risk measures such as VaR and CVaR.

Finally, the descriptive statistics for the explanatory variables support the empirical design used to test H3 and H4. Brent returns and carbon-price returns display substantial volatility and non-normality, indicating that energy and carbon markets experienced sizeable shocks over the sample period. Likewise, aggregate market returns in

Table 3

Orthogonalized factor determinants of the transition-related equity premium.

Variable	Europe	United States	United Kingdom
Market return	−0.1690*** (0.0313)	−0.0270 (0.0330)	−0.3408*** (0.0473)
Brent return	−0.1542*** (0.0171)	−0.2591*** (0.0243)	−0.1682*** (0.0199)
Carbon-price return	−0.0184** (0.0079)	−0.0178** (0.0088)	−0.0207** (0.0098)
Financial/stress factor, orthogonalized	0.0006 (0.0005)	0.0007 (0.0005)	0.0005 (0.0006)
Industrial factor, orthogonalized	0.0046*** (0.0004)	0.0024*** (0.0004)	0.0055*** (0.0004)
Observations	4230	4230	4230
Adj. R ²	0.135	0.128	0.146

Source: Authors' computation

Europe, the United States, and the United Kingdom exhibit negative skewness and excess kurtosis, consistent with episodes of stress and sharp repricing. Taken together, these characteristics support the subsequent modeling strategy, in which the transition spread is related to energy-price movements, carbon-pricing dynamics, and broad market conditions.

5.2. Baseline results

Table 1 reports the baseline OLS estimates for the determinants of the transition spread across the European, US, and UK equity markets. The results show that broad market conditions remain relevant, although their strength differs across markets. The coefficient on market returns is negative and statistically significant in Europe and the United Kingdom, indicating that the clean-minus-brown spread tends to narrow when aggregate equity markets rise in these two markets. The US coefficient has the same negative sign but is not statistically significant in the baseline specification. Overall, the evidence provides support for H4, while also pointing to cross-market heterogeneity in the strength of the market channel.

The role of transition-specific factors is also heterogeneous. Carbon-price returns enter with a negative coefficient in all three markets, but the effect is only weakly significant in Europe in the baseline specification. This suggests that carbon-pricing dynamics are associated with movements in the transition spread, but that their unconditional average effect is not uniformly strong across markets. As shown later in the extended determinant and subsample analyses, the carbon-pricing channel becomes more visible once broader industrial conditions and regime variation are taken into account.

Brent returns display a negative and statistically significant coefficient in all three regressions. This indicates that oil-price movements are systematically associated with a narrowing of the clean-minus-brown spread in the baseline model. Economically, this pattern is consistent with the idea that energy-market conditions affect the relative performance of clean and brown equity exposures, particularly because fossil-fuel-related sectors are directly linked to oil-market dynamics. The evidence therefore provides stronger support for the energy-price component of H3 than in the earlier unconditional descriptive evidence.

A further notable feature of Table 1 is the moderate explanatory power of the baseline regressions, with adjusted R² values of 0.058 in

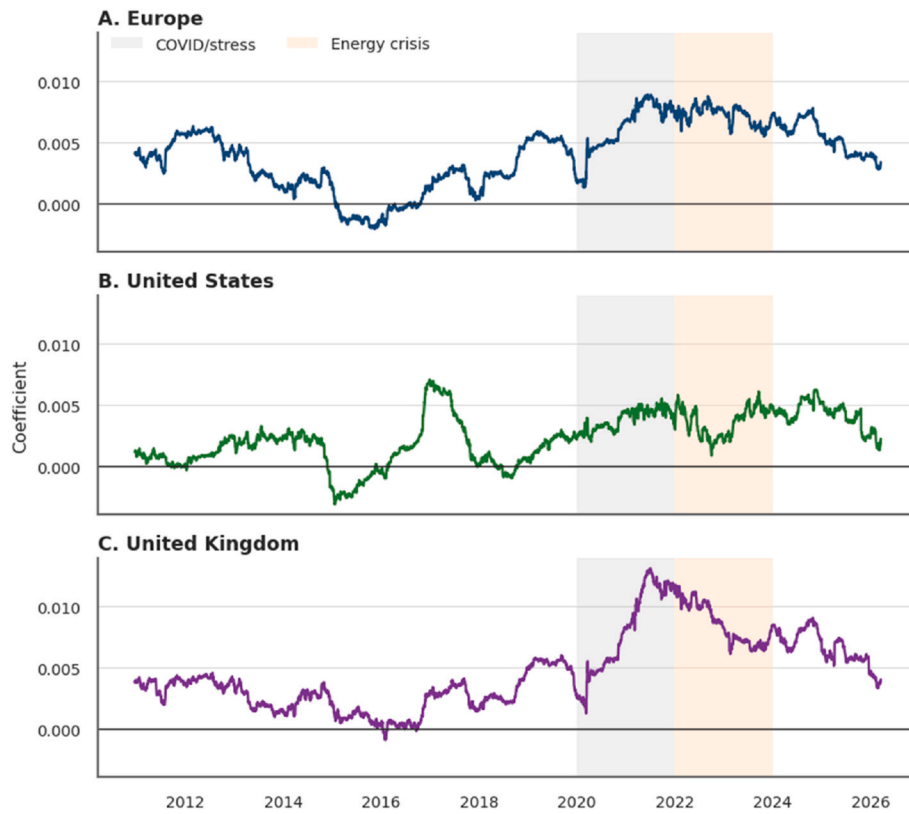


Fig. 2. Time-varying industrial determinant of the transition-related equity premium
Source: Authors' computation.

Table 4
Downside beta - sensitivity to negative market returns.

Market	Downside Beta	Std. Error	t-statistic	p-value	N (downside days)
Europe	-0.3731***	(0.0684)	-5.4541	0.0000	1941
United States	-0.2708***	(0.0693)	-3.9064	0.0001	1851
United Kingdom	-0.5117***	(0.0774)	-6.6071	0.0000	1923

Notes: ***, **, * denote statistical significance at the 1%, 5%, and 10% levels, respectively. Robust standard errors are reported in parentheses.

Source: Authors' computation

Europe, 0.111 in the United States, and 0.061 in the United Kingdom. While a substantial share of daily variation remains unexplained, this is not unexpected in high-frequency financial data. More importantly, the baseline results suggest that transition-risk pricing is shaped by broad market, energy-market, and carbon-pricing conditions, but may also depend on nonlinearities, tail events, sectoral exposures, and regime-specific mechanisms that are not fully captured by a simple linear conditional-mean framework. In this sense, Table 1 establishes an informative baseline while motivating the factor-based, time-varying, and downside analyses that follow.

5.3. Factor-based determinants and time-varying effects

Table 2 shows that the determinant factor blocks add meaningful explanatory power beyond the baseline specification. The improvement is especially clear for the industrial/real-economy factor. In Europe, the adjusted R^2 increases from 0.058 in the baseline model to 0.135 when the industrial factor is added, corresponding to an incremental gain of 0.077. A similar pattern is observed in the United Kingdom, where the

adjusted R^2 rises from 0.061 to 0.146, with an incremental gain of 0.085. The effect is smaller in the United States, but still positive: the adjusted R^2 increases from 0.111 to 0.127 when the industrial factor is included.

By contrast, the financial/stress factor adds more modest explanatory power. The increase in adjusted R^2 is 0.009 in Europe, 0.005 in the United States, and 0.028 in the United Kingdom. This suggests that financial stress contributes to variation in the transition spread, but less consistently than the industrial factor. The joint p-values below 0.001 across all three markets indicate that the added determinant blocks are jointly statistically significant. Overall, Table 2 supports the view that the transition-related equity premium is shaped not only by market, Brent, and carbon-price controls, but also by broader sectoral and real-economy conditions, with the industrial block providing the strongest incremental explanatory contribution.

Table 3 further tests whether the industrial and financial/stress factors contain information beyond broad market movements. To do so, both factor variables are orthogonalized with respect to the corresponding regional market return before entering the regressions. The results confirm that the industrial factor remains positive and highly statistically significant in all three markets. The coefficient is 0.0046 in Europe, 0.0024 in the United States, and 0.0055 in the United Kingdom, with significance at the 1% level throughout. This indicates that the role of industrial and real-economy conditions is not merely a restatement of aggregate market exposure, but captures an additional determinant of the clean-minus-brown spread.

By contrast, the orthogonalized financial/stress factor is not statistically significant in any of the three markets. This does not imply that financial stress is irrelevant, but rather that its independent contribution is weaker once broad market returns, Brent returns, carbon-price returns, and industrial conditions are jointly controlled for. This result is consistent with the interpretation that financial stress plays a more episodic or regime-dependent role, whereas industrial conditions represent the more robust determinant block.

Table 5
Quantile regression coefficients.

Market	Variable	Q10	Q25	Q50	Q75	Q90
Europe	Market return	-0.2524	-0.2553	-0.3154	-0.3564	-0.3239
	Brent oil return	-0.0015	0.0045	0.0000	-0.0242	-0.0410
	Carbon (EUA) return	-0.0287	-0.0231	-0.0234	-0.0427	-0.0615
United States	Market return	-0.1600	-0.1493	-0.1631	-0.2080	-0.2215
	Brent oil return	0.0166	-0.0076	-0.0035	-0.0208	-0.0235
	Carbon (EUA) return	-0.0318	-0.0192	-0.0205	-0.0280	-0.0152
United Kingdom	Market return	-0.3088	-0.3847	-0.4234	-0.4783	-0.5526
	Brent oil return	0.0378	0.0133	-0.0039	-0.0174	-0.0567
	Carbon (EUA) return	-0.0067	-0.0112	-0.0141	-0.0259	-0.0128

Notes: Reported coefficients are estimated from quantile regressions at the 10th, 25th, 50th, 75th, and 90th percentiles of the conditional distribution of the transition spread.

Source: Authors' computation

The remaining controls also provide useful information. Brent returns remain negative and statistically significant across all markets, while carbon-price returns become negative and significant in all three specifications. This suggests that energy-market and carbon-pricing dynamics continue to matter after controlling for market movements and the additional factor structure. Overall, Table 3 strengthens the interpretation that the transition-related equity premium is shaped by a combination of market, energy, carbon-pricing, and industrial conditions, with the industrial factor providing the most robust additional channel.

Fig. 2 complements these cross-sectional regressions by showing the time-varying contribution of the industrial factor. The figure reports 252-trading-day rolling coefficients for Europe, the United States, and the United Kingdom. Across the three markets, the industrial coefficient is generally positive, but its magnitude varies substantially over time. The effect becomes more pronounced during periods of macro-financial and energy-market stress, especially during the COVID/stress and energy-crisis windows. This pattern reinforces the view that the transition-related equity premium is not stable over time, but depends on broader industrial and real-economy conditions that vary across regimes.

Taken together, Tables 2 and 3 and Fig. 2 directly address the determinant-variable and time-varying dimensions of the analysis. They show that the transition spread is not only related to the baseline market, Brent, and carbon-price controls, but is also shaped by broader industrial conditions. The evidence therefore supports the main interpretation of the paper: the transition-related equity premium is conditional, heterogeneous, and regime-dependent rather than an unconditional average return differential.

Additional evidence on the construction and stability of the factor-based determinants is reported in Appendix A3. Table A3.1 shows that the financial/stress factor loads negatively on European and US financial-sector returns and positively on VSTOXX changes, consistent with its interpretation as a stress-oriented factor. The industrial factor, by contrast, loads positively on all industrials, materials, and utilities inputs, indicating that it captures a common real-economy sectoral component. Table A3.2 further shows that the industrial factor remains positive and statistically significant across all subsamples and markets, whereas the financial/stress factor displays a more episodic pattern. This appendix evidence reinforces the interpretation that industrial conditions are the most stable determinant block, while financial stress operates more conditionally across regimes.

5.4. Downside risk and regime analysis

5.4.1. Downside beta

Table 4 reports the downside beta estimates for the transition spread, obtained by conditioning the sample on days with negative aggregate market returns. The downside beta is negative and highly statistically

significant in all three markets, indicating that the relationship between broad market movements and the clean-minus-brown spread changes systematically under adverse market conditions. Since the downside regressions are estimated on negative market-return days, a negative coefficient implies that larger market declines are associated with an increase in the clean-minus-brown spread. In economic terms, this suggests that clean energy equities tend to perform relatively better than brown assets during downside market conditions. This result provides clear support for H6 and indicates that transition-risk pricing is asymmetric rather than constant across market environments.

The magnitude of the downside beta differs across markets, which is also informative. The strongest effect is observed in the United Kingdom, where the estimated downside beta is the most negative, followed by Europe and then the United States. This suggests that the UK transition spread is especially sensitive to market stress, a finding that is consistent with the earlier descriptive evidence showing stronger tail behavior and more extreme observations in the UK series. Europe also exhibits a substantial negative downside beta, indicating that the clean-minus-brown spread reacts strongly during negative market episodes. The US result, while smaller in magnitude, remains economically meaningful and highly significant, confirming that the asymmetry is not confined to a single jurisdiction.

These estimates add an important layer to the baseline OLS results. Whereas the baseline regressions in Table 1 show how the transition spread responds to market conditions on average, the downside beta analysis demonstrates that this relationship becomes particularly pronounced when market conditions deteriorate. This distinction is important. It suggests that the pricing of transition risk is not simply a linear average relationship, but is amplified during negative states of the world. Such a pattern is consistent with the broader interpretation that the transition-related equity premium is conditional and state-dependent.

More broadly, Table 4 also provides indirect support for H5 by showing that the transition spread is regime-dependent. Downside days can be understood as a specific market regime characterized by tighter financial conditions, elevated risk aversion, and stronger cross-asset repricing. The fact that the spread responds differently under these conditions indicates that transition-risk pricing is state-contingent. In that sense, the downside beta results bridge the gap between the baseline mean regressions and the more explicit regime-based analyses that follow, including quantile regressions, tail-risk metrics, and event studies.

Overall, Table 4 provides one of the clearest findings in the paper. Across all three markets, adverse market conditions are associated with a statistically and economically significant change in the transition spread. The evidence indicates that the clean-minus-brown spread rises during stronger market declines, implying relative clean-energy out-performance in downside states. The result is strongest in the UK, but is also clearly present in Europe and the United States, pointing to a

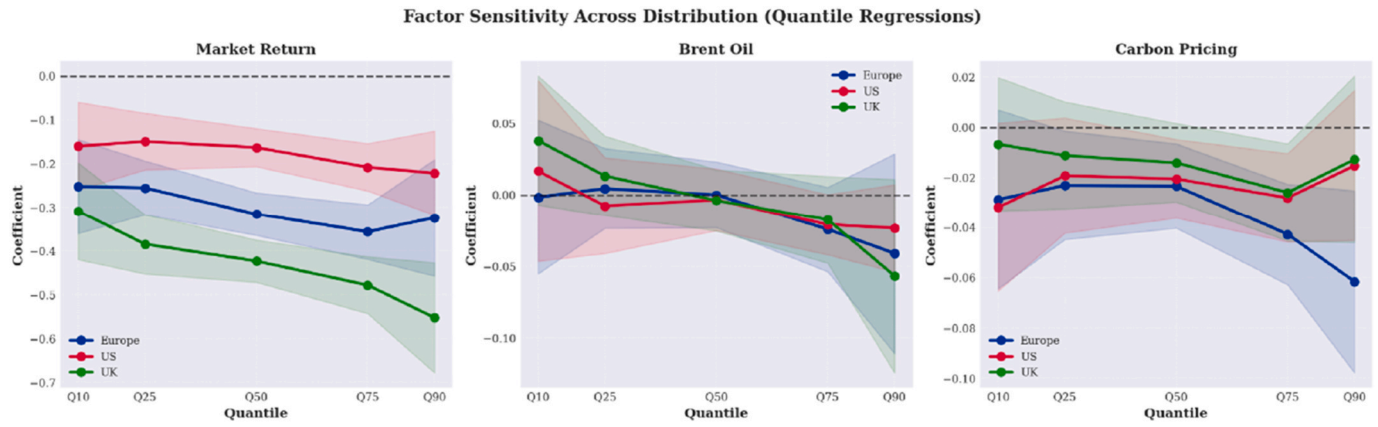


Fig. 3. Factor Sensitivity Across Distribution (Quantile Regressions)

Source: Authors' computation.

Table 6

Tail risk metrics.

Market	VaR (95%)	VaR (99%)	CVaR (95%)	CVaR (99%)
Europe	-0.0255	-0.0412	-0.0353	-0.0520
United Kingdom	-0.0268	-0.0477	-0.0414	-0.0700
United States	-0.0263	-0.0437	-0.0378	-0.0586

Source: Authors' computation

broadly shared asymmetry with heterogeneous intensity across markets.

5.4.2. Quantile Regressions

Table 5 reports the quantile regression estimates for the transition

spread across the European, US, and UK equity markets. Unlike the baseline OLS specification, which summarizes average effects, the quantile framework allows the sensitivity of the spread to vary across different parts of its conditional distribution. This is particularly relevant in the present setting, since the descriptive evidence reported in Appendix A2 indicates strong departures from normality and substantial tail behavior. The results suggest that the determinants of the transition spread are not constant across the conditional distribution, thereby providing further evidence in support of H6 and, more broadly, H5.

The most consistent result concerns the market-return coefficient, which is negative across all quantiles and all three markets. This indicates that the transition spread remains inversely related to aggregate market performance throughout the distribution, although the

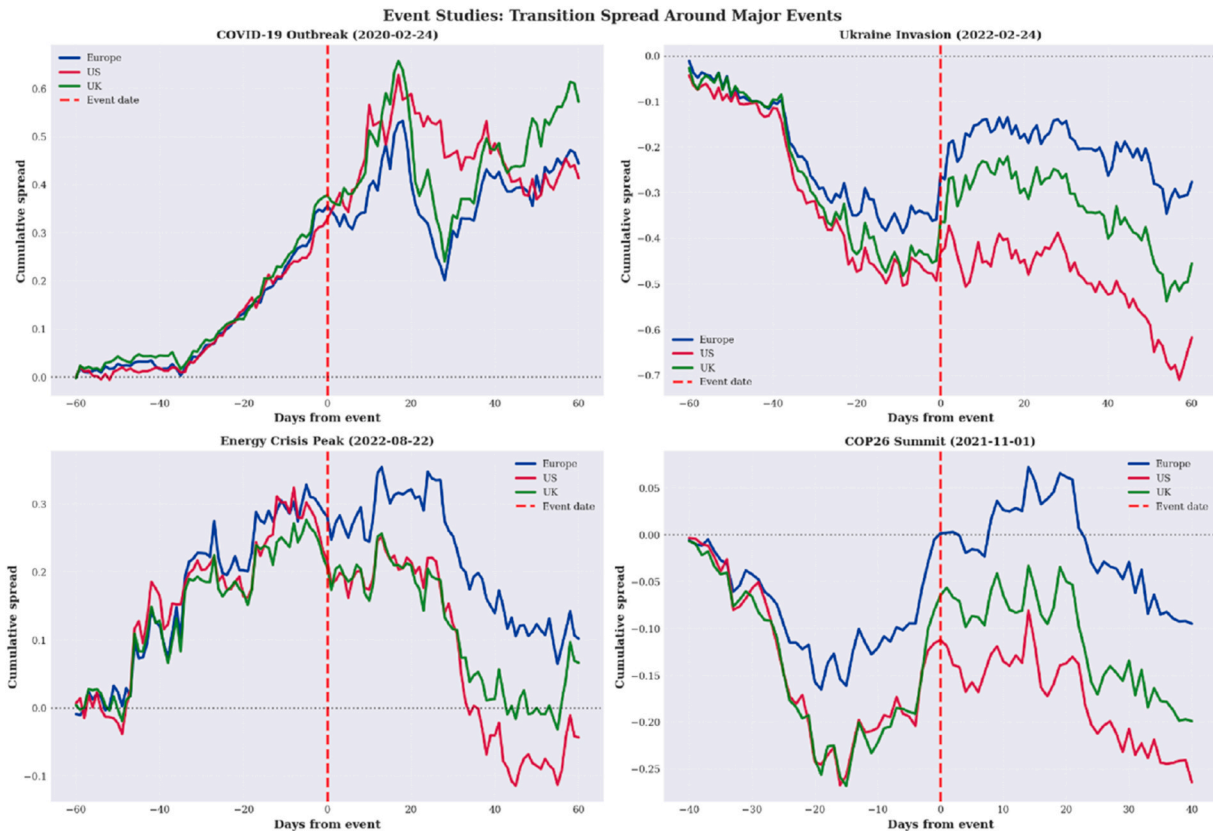


Fig. 4. Event Studies: Transition Spread Around Major Events

Source: Authors' computation.

Table 7

Cumulative spread (30 Days post-event).

Market	COP26 Summit	COVID-19 Outbreak	Energy Crisis Peak	Ukraine Invasion
Europe	-0.0291	0.3038	0.2528	-0.1427
UK	-0.1344	0.3362	0.1156	-0.2591
US	-0.2073	0.4646	0.1227	-0.4351

Notes: Values represent cumulative changes in the transition spread measured at the 30-day post-event horizon within each event window.

Source: Authors' computation

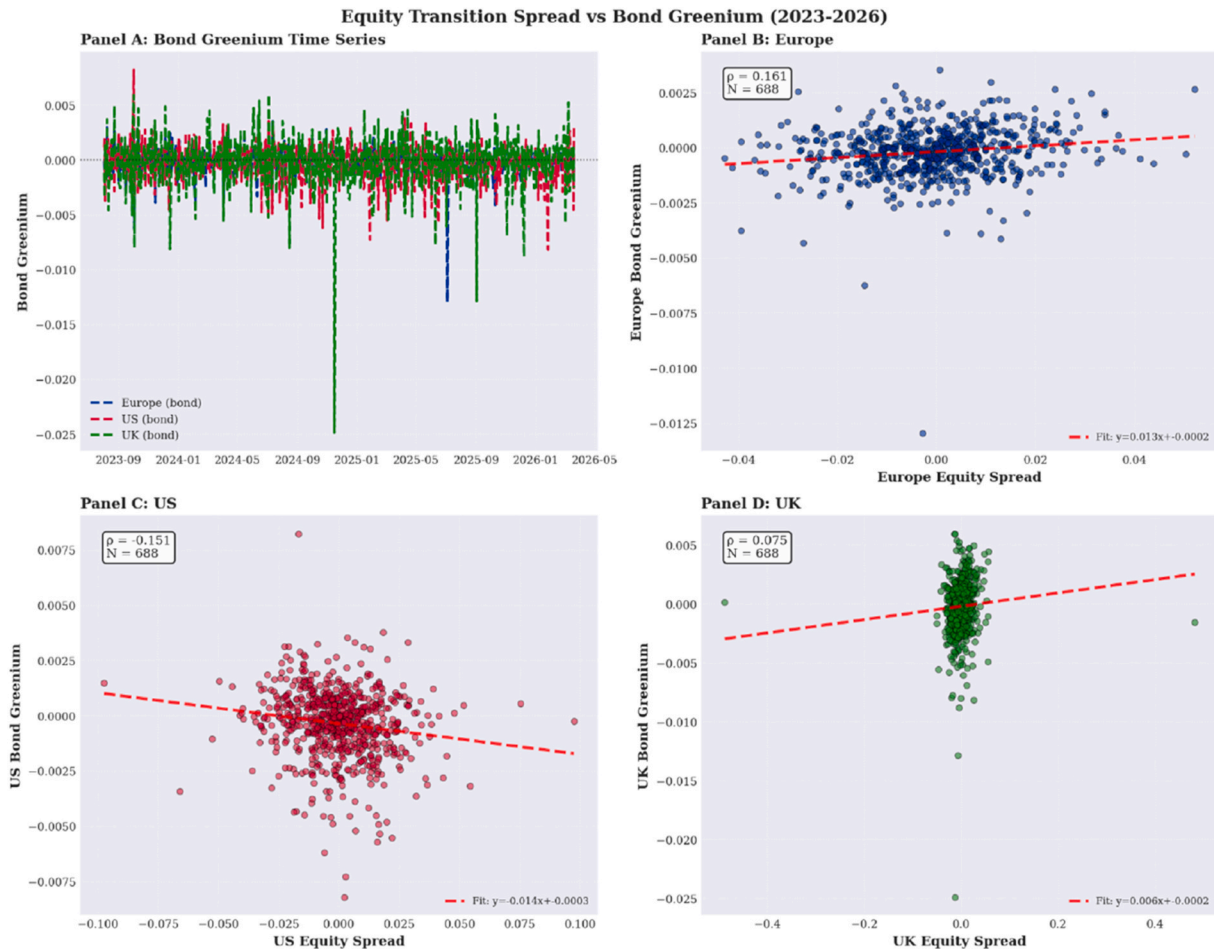
magnitude of this relationship varies. In Europe, the coefficient becomes more negative at higher quantiles, suggesting that market-wide conditions exert a stronger effect when the spread is relatively elevated. A similar pattern is visible in the United Kingdom, where the market coefficient becomes markedly more negative in the upper quantiles and reaches its largest absolute value at the 90th percentile. The US pattern is flatter, but the coefficient remains negative throughout. Taken together, these results indicate that the market effect identified in the baseline regressions is not uniform, but changes materially across the distribution of the spread.

The UK results are especially notable. Among the three markets, the United Kingdom exhibits the strongest quantile variation in the market-return coefficient, with the effect becoming substantially more negative in the upper tail. This suggests that periods of unusually high transition spreads are also periods in which the spread is more sensitive to broad market repricing. In other words, when the spread moves away from its

center, the UK market becomes particularly exposed to general market conditions. This pattern is consistent with the earlier evidence of greater volatility and fatter tails in the UK series and reinforces the interpretation that the UK market is characterized by stronger regime dependence and more pronounced tail sensitivity.

The Brent coefficient is less stable across quantiles than in the baseline OLS specification. In Europe, the oil effect is close to zero around the center of the distribution and becomes more negative in the upper tail. In the United States and the United Kingdom, the sign of the Brent coefficient changes across quantiles, suggesting that the effect of oil-price shocks is not uniform over the spread distribution. This does not contradict the baseline evidence in Table 1, where Brent returns are statistically significant across markets. Rather, it indicates that the average Brent effect is composed of heterogeneous quantile-specific responses. Oil-price dynamics therefore appear to matter systematically on average, but their influence varies across different states of the transition spread.

The carbon-price coefficient also displays meaningful state dependence. In Europe, the carbon effect is negative across all quantiles and becomes more negative toward the upper end of the distribution, reaching its largest magnitude at the 90th percentile. This suggests that carbon-pricing dynamics exert their strongest influence when the transition spread is relatively high. In the United States, the carbon coefficient also remains negative across all quantiles, although the variation is more moderate and the strongest effect appears in the lower tail. In the United Kingdom, the carbon coefficient is smaller in magnitude but generally remains negative. Overall, the quantile results suggest that

**Fig. 5.** Equity Transition Spread vs. Bond Greenium (2023 – 2026)

Source: Authors' computation.

carbon pricing is a state-dependent channel whose relevance varies across markets and across the distribution of the spread.

From a broader perspective, Table 5 shows that transition-risk pricing is conditional on the position of the spread within its own distribution. The economic relationships observed at the center do not necessarily carry over to the tails, and in several cases the slope coefficients become stronger at more extreme quantiles. This is precisely the type of pattern one would expect if the transition premium were asymmetric and regime-dependent rather than constant over time. For that reason, the quantile results complement the downside beta estimates in Table 4 and strengthen the conclusion that the transition spread behaves differently under adverse or unusual market conditions.

Overall, Table 5 provides evidence against a purely linear and homogeneous interpretation of transition-risk pricing. The market-return coefficient is consistently negative but quantile-dependent, especially in Europe and the United Kingdom. Brent and carbon-price returns also display distribution-dependent effects, indicating that energy-market and carbon-pricing channels operate differently across states of the spread. These findings reinforce the argument that the transition-related equity premium is shaped by asymmetry, tail sensitivity, and market-specific regimes, thereby offering substantive support for H5 and H6 and more nuanced support for H3.

Fig. 3 provides a visual summary of the quantile-regression estimates reported in Table 5 and confirms that the determinants of the transition spread vary across the conditional distribution. The most robust pattern concerns aggregate market returns, whose coefficients are negative at all quantiles and across all three markets. This indicates that the spread remains systematically linked to broad market conditions, although the strength of the relationship differs across states of the distribution. The effect is particularly pronounced in the United Kingdom and, to a lesser extent, in Europe, where the coefficient becomes more negative at higher quantiles.

The Brent coefficients display a more heterogeneous profile across quantiles and markets. While the baseline regressions indicate a significant average relationship between Brent returns and the transition spread, Fig. 3 shows that this relationship is not uniform across the conditional distribution. In some markets, the Brent effect is close to zero near the center of the distribution but becomes more negative in the upper quantiles, suggesting that oil-price dynamics may matter more in specific states of the spread than in others. Carbon-pricing coefficients are generally negative and also display state dependence, especially in Europe and the United States, where the effect becomes stronger in certain parts of the distribution. Overall, the figure reinforces the interpretation that transition-risk pricing is asymmetric and state-dependent, thereby supporting H5 and H6, while offering more nuanced support for H3 through energy-price and carbon-pricing channels.

5.4.3. Tail risk metrics

Table 6 reports the tail-risk properties of the transition spread through historical VaR and CVaR measures at the 95% and 99% confidence levels. The results show that all three markets are exposed to substantial downside tail risk, with losses becoming markedly larger in the most adverse scenarios. In each case, CVaR is more negative than VaR, indicating that once the spread enters the tail, losses tend to deepen beyond the threshold itself. This confirms that extreme downside outcomes are economically relevant and supports the view that transition-risk pricing is not adequately captured by average effects alone.

Cross-market differences are also evident. The United Kingdom exhibits the largest downside risk, with the most negative VaR and CVaR values at both confidence levels, followed by the United States and Europe. This pattern is consistent with the earlier evidence from the descriptive statistics and quantile analysis, which pointed to stronger tail sensitivity and greater instability in the UK spread. Overall, Table 6 reinforces the evidence of downside asymmetry and provides additional support for H6, while also fitting the broader interpretation of regime-

Table 8

Equity Spread vs. Bond Greenium.

Table 8A. Correlation between Equity Spread and Bond Greenium					
Market	Correlation	N	p-value		
Europe	0.1612	688	0.0000		
United States	−0.1512	688	0.0001		
United Kingdom	0.0754	688	0.0480		

Table 8B. Validation Regression (greenium _t = α + β · equity_spread _t)					
Market	Beta	Std. Error	t-statistic	p-value	R ²
Europe	0.0135	0.0032	4.2720	0.0000	0.0260
United States	−0.0139	0.0036	−3.8936	0.0001	0.0229
United Kingdom	0.0057	0.0063	0.8932	0.3717	0.0057

Notes: Table 8A reports Pearson correlations between the equity transition spread and the bond greenium. Table 8B reports OLS estimates of the relationship between bond greenium and equity spread.

dependent pricing under H5.

5.5. Event-based analysis

Fig. 4 examines the cumulative behavior of the transition spread around four major events: the COVID-19 outbreak, the Russian invasion of Ukraine, the 2022 energy crisis peak, and the COP26 summit. The figure shows that the transition spread responds differently across episodes and markets, indicating that transition-risk pricing is strongly regime-dependent. Rather than displaying a stable pattern over time, the spread widens or contracts depending on the nature of the underlying shock, providing clear support for H5.

The strongest spread deterioration is observed around the Ukraine invasion, where cumulative spreads turn sharply negative, especially in the United States and the United Kingdom, indicating pronounced brown outperformance during a severe energy-security shock. By contrast, the COVID-19 episode is associated with a positive cumulative spread path, suggesting a more favorable relative performance of clean energy assets in the subsequent window. The energy crisis and COP26 panels further confirm that market responses are heterogeneous: Europe appears comparatively more resilient in both episodes, while the US and UK markets display more persistent negative post-event dynamics. Overall, Fig. 4 reinforces the view that the transition premium is not constant, but instead depends critically on the broader macro-financial and policy environment.

Table 7 reports the 30-day cumulative spread following each major event and confirms that the response of the transition spread differs substantially across shock episodes and markets. The results provide clear evidence of regime dependence, in line with H5. In particular, the post-event spread is positive after the COVID-19 outbreak and the 2022 energy crisis peak in all three markets, indicating relative clean-energy outperformance over those windows. By contrast, the spread turns negative after the Ukraine invasion and after the COP26 summit, implying relative brown outperformance in the aftermath of those events.

Cross-market differences are also pronounced. The United States exhibits the strongest positive cumulative spread following COVID-19, but also the largest negative response after the Ukraine invasion, suggesting that the US market reacts more sharply to major global shocks. The United Kingdom records the most negative post-COP26 outcome, while Europe appears comparatively more resilient, with smaller cumulative declines in the adverse-event windows. Taken together, these findings reinforce the interpretation that transition-risk pricing is not stable over time, but varies materially with the nature of the underlying macro-financial or policy shock, thereby offering additional support for H5.

5.6. Robustness - bond greenium

Fig. 5 compares the equity transition spread with the bond greenium over the shorter 2023-2026 sample and provides a visual robustness check for H7. Panel A shows that the bond greenium fluctuates closely around zero in all three markets, with occasional negative spikes but no persistent trend. This suggests that green bond pricing differentials are present, but relatively small in daily return terms over the available sample. Panels B-D examine the contemporaneous relationship between the equity transition spread and the bond greenium in each market through scatter plots and fitted linear trends.

The visual evidence points to only limited cross-asset consistency. In Europe, the relationship appears weakly positive, suggesting that periods of wider equity transition spreads may coincide with slightly stronger bond greenium dynamics. In contrast, the United States displays a weak negative association, while the United Kingdom shows only a very modest positive slope combined with substantial dispersion. In all three cases, the point clouds are widely scattered, indicating that the contemporaneous relationship between equity and bond pricing signals is noisy and economically limited over the short sample.

Overall, Fig. 5 provides at most modest visual support for H7. While there is some indication that equity and bond measures of transition-related pricing may move in the same direction in Europe and, to a lesser extent, in the United Kingdom, the relationship is neither strong nor uniform across markets. The figure therefore suggests that any cross-asset consistency is partial and should be interpreted cautiously, especially given the short sample period and the fact that the equity spread and bond greenium are not constructed from perfectly analogous instruments.

Table 8 reports the robustness analysis linking the equity transition spread to the bond greenium over the shorter 2023-2026 sample. The results suggest that cross-asset consistency is present, but only partially and with important differences across markets. In the correlation analysis, Europe displays a positive and statistically significant association between the equity spread and the bond greenium, while the United Kingdom also shows a positive, though much smaller, correlation. By contrast, the United States exhibits a significant negative correlation. The regression results convey the same message: the estimated slope is positive and highly significant in Europe, negative and significant in the United States, and statistically insignificant in the United Kingdom. Taken together, the evidence does not support a uniform cross-market relationship between equity-based and bond-based transition pricing.

The European results provide the clearest support for H7. Both the positive correlation and the positive regression coefficient suggest that periods of stronger clean-versus-brown equity performance tend to coincide with a more favorable pricing differential for green bonds relative to conventional corporate bonds. This pattern is consistent with the idea that transition-related pricing may extend across asset classes in Europe, where sustainable finance markets and carbon-related pricing signals are likely to be more deeply embedded. In contrast, the US findings point in the opposite direction, indicating that equity and bond measures may capture different dimensions of transition-related pricing over the short sample. For the United Kingdom, the evidence is weaker, as the correlation is only modest and the regression coefficient is not statistically different from zero.

Overall, Table 8 suggests that H7 receives only partial support. The results are consistent with cross-asset alignment in Europe, but this pattern is neither replicated in the United States nor strongly confirmed in the United Kingdom. Accordingly, the robustness exercise should be interpreted with caution. The short sample period limits statistical power, and the two pricing measures are not perfectly analogous, since the equity spread captures the relative performance of clean and brown stocks, whereas the bond greenium reflects the pricing differential between green and conventional bond instruments. Even so, the European evidence is informative and suggests that the transition-related pricing signal identified in equity markets may, at least in some cases, extend to

fixed-income markets as well.

A concise summary of the hypothesis-testing outcomes is reported in Appendix A1.

6. Conclusions

This paper has examined the pricing of climate transition risk in equity markets through the return differential between clean energy and brown equity indices across Europe, the United States, and the United Kingdom. Using daily data and a range of complementary empirical approaches, the analysis shows that the transition-related equity premium is not well described as a large, stable, unconditional spread. Rather, it emerges as a conditional and heterogeneous phenomenon, shaped by broad market conditions, energy-market and carbon-pricing dynamics, industrial and real-economy conditions, and periods of stress. In this respect, the findings point to a form of transition-risk pricing that is episodic, nonlinear, and closely tied to market conditions.

A central result of the paper is that unconditional average spreads are economically small, offering only limited support for a strong mean-based transition premium. Yet this does not imply that transition risk is absent from asset prices. On the contrary, the evidence shows that the spread varies systematically over time and across markets, with particularly pronounced differences in volatility and tail behavior. The United Kingdom stands out as the market most exposed to extreme spread realizations, while Europe and the United States display more moderate, though still clearly non-normal, distributions. These differences suggest that the pricing of transition risk is not uniform across jurisdictions, but instead reflects variation in market structure, sectoral exposure, and institutional context.

The multivariate results indicate that the transition spread is shaped by several determinant channels. In the baseline specification, market returns, Brent returns, and carbon-price returns are associated with movements in the clean-minus-brown spread, although their statistical strength differs across markets. The extended factor-based analysis adds an important layer to this interpretation. The industrial/real-economy factor provides the strongest incremental explanatory contribution and remains statistically significant even after being orthogonalized with respect to regional market returns. This suggests that industrial conditions are not merely a restatement of broad market exposure, but represent a distinct determinant of the transition-related equity premium. By contrast, the financial/stress factor plays a more episodic role, contributing in some specifications and subsamples but not retaining a robust independent effect in the full orthogonalized model.

The most distinctive contribution of the paper lies in the evidence on asymmetry and regime dependence. The downside-beta results show that the transition spread responds significantly during negative market conditions, indicating that the relationship between market conditions and relative clean-versus-brown performance changes when markets deteriorate. Given the clean-minus-brown definition of the spread, the negative downside-beta estimates imply that stronger market declines are associated with a higher transition spread, suggesting relative clean-energy outperformance in downside states. The quantile regressions reinforce the broader conclusion that the sensitivity of the spread to market, energy, and carbon-related factors is not constant across its distribution, but varies materially between central and tail realizations. The tail-risk measures point in the same direction, showing that extreme downside spread outcomes are economically meaningful and especially pronounced in the UK market. Event-based evidence likewise shows that spread dynamics differ substantially across major episodes such as the COVID-19 outbreak, the invasion of Ukraine, the energy crisis, and COP26. Taken together, these results indicate that transition-risk pricing is fundamentally state-contingent and becomes most visible during periods of market stress or elevated uncertainty.

The bond-market robustness exercise yields a more qualified picture. Evidence of cross-asset consistency is strongest in Europe, where the equity transition spread is positively related to the bond greenium

proxy, suggesting that transition-related pricing may extend across the capital structure in that market. By contrast, the relationship is weaker in the United Kingdom and negative in the United States, indicating that the alignment between equity- and bond-based measures of transition pricing is far from universal. The robustness results therefore suggest that cross-asset consistency is partial rather than general and likely depends on differences in market design, investor base, and the maturity of sustainable finance ecosystems.

These findings carry several implications for policy and financial supervision. First, the state-dependent nature of the transition spread suggests that climate-related financial vulnerabilities are unlikely to be fully captured by unconditional averages or static risk metrics. Supervisory frameworks should therefore incorporate downside-sensitive and scenario-based approaches when assessing climate-related market exposures. Second, the relevance of carbon-pricing and energy-market dynamics indicates that policy and commodity-market signals can shape relative green-versus-brown equity valuations in meaningful ways. Credible and persistent climate-policy frameworks may thus influence financial valuations not only through expected cash-flow effects, but also through the relative repricing of green and brown sectors. Third, the importance of industrial and real-economy conditions suggests that transition-risk pricing is closely connected to sectoral structure and the broader macro-industrial environment. This underlines the need to align transition policy with energy-security, industrial-policy, and financial-stability objectives.

The paper is subject to a number of limitations. The analysis relies on broad index-level proxies for green and brown assets, which necessarily abstract from substantial firm-level heterogeneity. The event-based analysis traces cumulative spread behavior around major shocks but does not constitute a fully model-based abnormal-return framework and should therefore be interpreted with caution. The bond-market exercise is constrained by the short sample period, limiting statistical power and making the results more sensitive to short-run fluctuations. In addition, while the empirical specification includes market returns, oil prices, carbon prices, and factor-based measures of financial and industrial conditions, other potentially relevant variables, such as interest rates,

inflation surprises, firm-level fundamentals, or policy announcements, are not modeled explicitly. Finally, the use of daily data is well suited to identifying high-frequency repricing but may also obscure slower-moving structural channels through which transition risk enters asset prices.

These limitations suggest several fruitful directions for future research. One natural extension would be to move from index-level to firm-level or sector-level evidence in order to identify more precisely how transition exposure, energy dependence, and technological positioning affect relative pricing. Another would be to incorporate more granular climate-policy measures, including regulatory announcements, carbon-market reforms, and green industrial policies, in order to distinguish more clearly between policy and market-driven channels. Further work could also adopt more formal nonlinear or regime-switching frameworks to model the state-contingent behavior of the transition premium directly. Finally, the cross-asset dimension deserves deeper attention, particularly through the analysis of credit spreads, sovereign bonds, derivatives, and fund flows, which may reveal additional channels through which transition risk is transmitted across financial markets.

Overall, the evidence presented in this paper suggests that climate transition risk is reflected in equity prices, but in a manner that is conditional, asymmetric, and heterogeneous across markets. The transition spread is not simply a stable premium earned in normal times; rather, it is a market- and regime-dependent pricing differential that becomes especially salient during stress periods and in the tails of the distribution. By documenting these features across three major markets and linking them to market-wide, energy-market, carbon-pricing, and industrial determinants, the paper contributes to a more nuanced understanding of how the low-carbon transition is priced in financial markets.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A1. Summary of Hypothesis Testing

Hypothesis	Main evidence	Conclusion
H1	Table A2.1; Fig. 1	Partially supported. Average spreads are small, but clearly time-varying.
H2	Table A2.1; Fig. 1	Supported. Spread dynamics differ across Europe, the US, and the UK.
H3	Tables 1, 3, and 5; Fig. 3	Partially supported. Carbon pricing matters more consistently than oil prices.
H4	Tables 1 and 3; Fig. 3	Strongly supported. Market-wide conditions are a robust determinant of the spread.
H5	Tables 4–7; Figs. 2–4	Supported. Spread behavior varies across major shock episodes and market regimes.
H6	Tables 4–6; Fig. 3	Strongly supported. The spread exhibits downside asymmetry and significant tail sensitivity.
H7	Table 8; Fig. 5	Partially supported. Cross-asset consistency is visible in Europe, but not uniformly across markets.

Appendix A2. Descriptive Statistics and Distributional Diagnostics

Table A2.1

Summary Statistics of Daily Returns

Series	N	Mean	Std. Dev.	Min	Q25	Median	Q75	Max	Skew	Kurtosis	JB p-value
Spread: Europe	4230	−0.0001	0.0162	−0.0901	−0.0095	−0.0002	0.0090	0.0945	0.1672	2.3486	0.0000
Spread: US	4230	−0.0002	0.0164	−0.1323	−0.0089	0.0000	0.0086	0.1130	−0.0736	4.2209	0.0000
Spread: UK	4230	−0.0001	0.0203	−0.4879	−0.0096	−0.0000	0.0086	0.4818	−0.0493	154.8893	0.0000
Market: Europe	4230	0.0002	0.0102	−0.1219	−0.0043	0.0005	0.0054	0.0807	−0.8096	10.0477	0.0000
Market: US	4230	0.0004	0.0107	−0.1277	−0.0036	0.0004	0.0054	0.0909	−0.6182	14.1684	0.0000
Market: UK	4230	0.0001	0.0096	−0.1151	−0.0043	0.0003	0.0050	0.0867	−0.7057	10.2998	0.0000
Brent Oil	4230	0.0001	0.0213	−0.4078	0.0000	0.0000	0.0000	0.3238	−1.9589	84.0448	0.0000
Carbon (EUA)	4230	0.0004	0.0296	−0.4321	−0.0135	0.0000	0.0156	0.2452	−0.7547	16.3116	0.0000

Source: Authors' computation

Distribution Properties of Transition Spreads

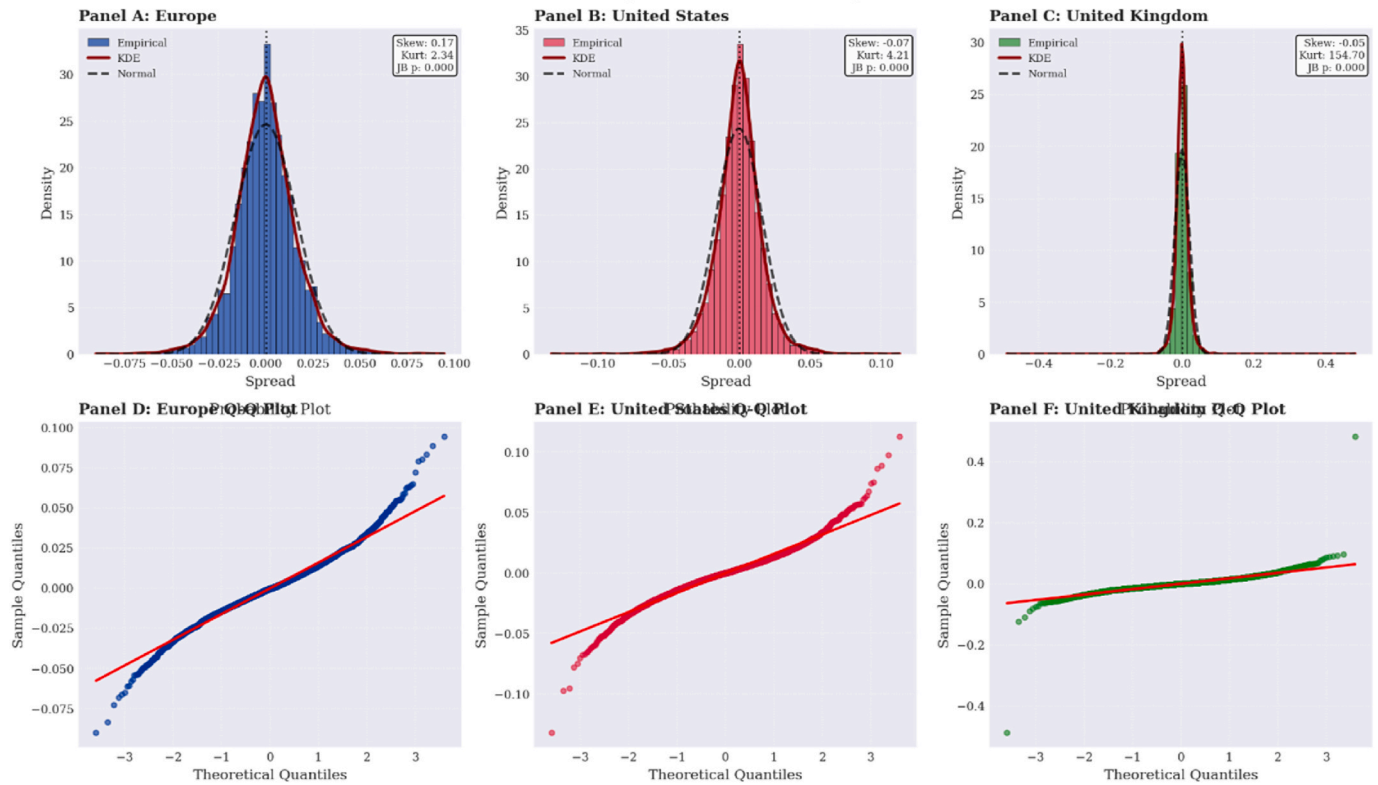


Fig. A2.2. Distribution properties of transition spreads
Source: Authors' computation.

Appendix A3. Factor Construction and Subsample Determinant Analysis

Table A3.1
Principal-component factor loadings

Input	Financial/stress factor	Industrial factor
r_fin_eu	-0.620	
r_fin_us	-0.541	
vstoxx_ret	0.568	
r_ind_eu		0.437
r_ind_us		0.438
r_mat_eu		0.431
r_mat_us		0.437
r_util_eu		0.376
r_util_us		0.316

Source: Authors' computation

Table A3.2
Subsample determinant regressions

Period	Market	Brent return	Carbon return	Financial/stress factor	Industrial factor
COVID/stress	Europe	-0.1051**	-0.0111	0.0053***	0.0062***
COVID/stress	United Kingdom	-0.1385***	-0.0398	0.0047***	0.0082***
COVID/stress	United States	-0.1986***	-0.0524	0.0062***	0.0034***
Energy crisis	Europe	-0.2634***	-0.1057***	0.0036**	0.0073***
Energy crisis	United Kingdom	-0.2545***	-0.0820**	0.0015	0.0081***
Energy crisis	United States	-0.3388***	-0.1047**	0.0027*	0.0035***
Post-crisis	Europe	-0.1686***	0.0068	0.0009	0.0043***
Post-crisis	United Kingdom	-0.1982***	0.0470	0.0032	0.0052***
Post-crisis	United States	-0.2895**	0.0344	0.0006	0.0032***
Pre-COVID	Europe	-0.1390***	-0.0083	-0.0018***	0.0032***
Pre-COVID	United Kingdom	-0.1292***	-0.0113	-0.0017***	0.0036***
Pre-COVID	United States	-0.2272***	-0.0035	-0.0018***	0.0021***

Source: Authors' computation

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